

A Report on
AN ADVANCED SECURITY FEATURE OF
MOTORCYCLE LOCKING SYSTEM BY USING
BIOMETRIC SCANNER

Submitted in the partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

In

ELECTRONICS AND COMMUNICATION ENGINEERING

By

M.Keerthana
(20NM1A04A5)
K. Surya Prasanna
(20NM1A0478)

Neha Kumari
(20NM1A04B9)
P. Yogitha
(20NM1A04C4)

Under the Esteemed Guidance of

Mr. B. Hari Kumar
Assistant Professor
Department of ECE
VIEW, Visakhapatnam



Department of Electronics and Communication
Engineering

VIGNAN'S INSTITUTE OF ENGINEERING FOR WOMEN

(Approved by AICTE, Affiliated to JNT University -GV, Vizianagaram)
Kapujaggarajupeta , VSEZ (P.O), Visakhapatnam -530 049 .A.P

2023-24

VIGNAN'S INSTITUTE OF ENGINEERING FOR WOMEN

(Approved by AICTE, Affiliated to JNT University -GV, Vizianagaram)

Kapujaggarajupeta , VSEZ (P.O), Visakhapatnam -530 049 .A.P

December 2023



CERTIFICATE

This is to certify that the report entitled “**AN ADVANCED SECURITY FEATURE OF MOTORCYCLE LOCKING SYSTEM BY USING BIOMETRIC SCANNER**” that is being submitted by

M.KEERTHANA

(20NM1A04A5)

NEHA KUMARI

(20NM1A04B9)

K. SURYA PRASANNA

(20NM1A0478)

P. YOGITHA

(20NM1A04C4)

in the partial fulfillment for the award of **Bachelor of Technology in Electronics and Communication Engineering** during 2023-24, in Vignan's Institute of Engineering for Women, affiliated to JNTU-GV University, is a record of bonafide work carried out by them under our guidance and supervision. The results embodied in this work have not been submitted to any other University or Institute for the award of any degree or diploma.

Signature of the Project Guide

Mr. B. Hari Kumar

Assistant Professor

Department of ECE

VIEW

Signature of the Head of the Department

Dr. D. Venkata Ramana

Associate Professor & HoD

Department of ECE

VIEW

External Examiner

DECLARATION

We hereby declare that the work entitled “**AN ADVANCED SECURITY FEATURE OF MOTORCYCLE LOCKING SYSTEM BY USING BIOMETRIC SCANNER**” is being submitted to **Vignan’s Institute of Engineering for Women**, Affiliated to **JNTU-GV** University, in partial fulfillment for the award of B.Tech degree during the year 2020-2024 in **Electronics and Communication Engineering**. The work was originally designed and executed by us under the guidance of our supervisor **Mr.B.Hari Kumar** and was not a duplication of work done by someone else. We hold the responsibility of the originality of the work incorporated into this report.

Signature of the candidate

Name: **M.KEERTHANA**

Date:

Signature of the candidate:

Name: **NEHA KUMARI**

Date:

Signature of the candidate:

Name: **K. SURYA PRASANNA**

Date:

Signature of the candidate:

Name: **P.YOGITHA**

Date:

ABSTRACT

This fingerprint system is a project to prevent motorcycle theft these days. So with our project or initiative, this will reduce the risk of thefts that happen. We got this idea based on the collected problem statement, which is that nowadays motorcycles are easily stolen by the thieves. The objective of this project is to upgrade the safety system, ease the ignition of the bike, and reduce the loss of motorcycles in the present time. Basically, we are making a new way to start a bike that was originally used as a starter. This Biometric Security will act as an additional security feature for our Motorcycle. By using our project, they can ignite their motorcycle just by using their fingerprint. This way is safer because the owner can only use their own bike. The Methodology used in this research is: first, we make some questionnaires and distribute them to motorcyclists because this will help us get supportive feedback about their motorcycle's safety features. In our preliminary study, this project requires us to study electronic devices and circuits. Thus, we have to do research to make a circuit so the fingerprint sensor will work well. In conclusion, we hope our project could decrease the rate of stolen motorcycles and provide a new interesting way of starting our motorcycle.

Keywords: Fingerprint sensor, Biometric security system, Anti-theft system, Keyless ignition, Arduino uno , Buzzer.

CONTENTS

Chapter no.	Title	Page no.
	Cover page	
	Certification	
	Declaration	
	Abstract	i
	Contents	ii
	List of Figures	iii
 Chapter 1	 Introduction	 1-3
	1.1 Introduction	1
	1.2 Motivation	1
	1.3 Area of Utility	2
	1.4 Literature Survey	2
	1.5 Objectives	3
	1.6 Advantages of Proposed	3
	1.7 Organization of Thesis	3
 Chapter 2	 Introduction of Designer Tools	 4-8
	2.1 Arduino IDE	4
	2.2 Proteus software	6
 Chapter 3	 Introduction of Hardware components	 9-18
	3.1 Arduino UNO	9
	3.2 Finger print sensor	12
	3.3 Relay Module	14
	3.4 Buzzer	14
	3.5 Battery	16
	3.6 Jumper Cables	17

Chapter 4	Implementation of Hardware and Software components	19-31
	4.1 Working	19
	4.2 Flow Chart	20
	4.3 Program Representation	21
	4.4 Circuit Connection Procedure	31
Chapter 5	Conclusion	32-34
	5.1 Enrolling Fingerprint using serial monitor	32
	5.2 Detection of valid/Invalid Fingerprint	33
	References	35

LIST OF FIGURES

Figure 2.1	: Arduino IDE software.....	3
Figure 2.2	: Proteus Software.....	4
Figure 3.1.1:	Pin Diagram of Arduino Uno Board.....	6
Figure 3.1.2:	Arduino Uno Board.....	7
Figure 3.2	: R307 Fingerprint Sensor.....	8
Figure 3.3	: Relay Module.....	8
Figure 3.4	: Buzzer.....	9
Figure 3.5	: Battery.....	10
Figure 3.6	: Jumper Cables.....	11
Figure 4.1	: Block Diagram of Proposed Model.....	12
Figure 4.2	: Flowchart.....	13
Figure 4.4	: Circuit Implementation in Proteus Software	24
Figure 5.1.1:	Adding of Adafruit Fingerprint library.....	25
Figure 5.1.2:	Enrolling the fingerprint using serial monitor.....	26
Figure 5.2	: Detection of valid/Invalid fingerprint using serial monitor.....	26

CHAPTER 1

INTRODUCTION

1.1. INTRODUCTION

In this project, We will be interfacing the fingerprint sensor with an Arduino Uno to design an Advanced Security Feature for a Motorcycle Locking System using Biometric Scanner .Security is a major concern in our day-to-day lives, and digital locks have become an important part of these security systems. The Biometric Scanner can be used to identify authorized users of a motorcycle by scanning their fingerprints. This Biometric Scanner will provide enhanced security as an additional layer, convenience, and versatility where it can be integrated with variety of other security feature such as alarm or buzzers.

Fingerprint sensor based is one of the safest motorcycle locking and starting system as it has A ability to identify and distinguish every person individually without making an error. This Module is very small that it can be kept anywhere, and with the portability feature and less power consumption, you can carry it to any place as well.

In Conclusion, This project proposes an Advanced Security Feature for Motorcycle Systems by incorporating a Biometric Scanner for user Authentication. Traditional Motorcycle Security relies on Keys and Ignition Systems, but these can be vulnerable to theft or unauthorized use. The integration of Biometric Scanner adds an extra layer of production, ensuring that only pre-registered and authorized users can start the Motorcycle.

1.2 MOTIVATION

Vehicle security is an important issue these days due to the rising number of vehicle thefts. Also, one more issue with vehicles is handling their keys. Keys need to be carried, and misplacing or losing them will cause a serious issue. Here we propose a solution to this problem by using a fingerprint-authenticated vehicle starter system. The system provides s secure and hassle-free way to start or stop the vehicle's engine. The user just needs to scan a finger to start the car; there is no need to carry any keys. The system only allows authorized users to start the vehicle.Users can first register with the system be scanning their fingerprints. The

system allows multiple users to register as authorized users. When in monitoring mode, the system checks for users to scan. On scanning, the system checks if the user is an authorized user and starts the vehicle for authorized users only. This system automates vehicle security using a fingerprint based system.

1.3. AREA OF UTILITY

The primary focus of this project is to prevent vehicles from being theft. The Fingerprint is unique for each and everyone. So protection of the vehicles can be done. It can be used in cars, bikes, and even in the heavy vehicles.

1.4. LITERATURE SURVEY

1.4.1. Existing system

In present-day automobiles, a high-voltage pulse is directly applied to coils that sit on top of the spark plugs (known as coil-on-plug). The major components of these systems are a coil pack, an ignition module, a crankshaft reluctor ring, a magnetic sensor, and an electronic control module. The ignition module controls the primary circuit to the coils, turning them on and off. The reluctor ring is mounted on the crankshaft so that as the crankshaft rotates, the magnetic sensor is triggered by notches in the reluctor ring. The magnetic sensor provides position information to the electronic control module, which governs ignition timing.

1.4.2. Proposed system

We developed a way to control the ignition of the vehicle through the fingerprint scanner. This system consists of the Arduino, which is the microcontroller for the project. To make sure the system is secure, only an authorized fingerprint is paired with the Arduino to start the ignition. Vehicles ignite when the enrolled fingerprint is matched against the fingerprints in the database, while users with no match in the database are prevented from igniting the vehicle. A theft alarm from Buzzer, a notification to the owner's mobile phone via SMS, and status are the appropriate signals to the owner. This article describes briefly and in detail the design and implementation of the ignition system.

1.5. OBJECTIVES

- The primary objective of this project is to prevent vehicle theft. To develop biometric based suitable vehicle security system. This Anti- theft fingerprint locking system is one of the advanced security systems.
- To develop a biometric system that provides a secure and hassle freeway to start the vehicle Engine and the system only allow authorized users to start the vehicle.

1.6. ADVANTAGES OF PROPOSED

- Highly efficient
- Enables automation in vehicles
- Avoid theft of vehicles
- System is flexible

1.7. ORGANIZATION OF THESIS

Chapter 1 In this chapter, thesis covers Introduction, Motivation and Literature.

Chapter 2 This chapter includes the Introduction of Arduino IDE, Introduction of Proteus Software.

Chapter 3 This chapter gives the brief discussion of the hardware components which are used in this project.

Chapter 4 This chapter shows the block diagram, working and Implementation of software and hardware components of the project.

Chapter 5 This chapter describes the results, conclusion and the References of this project.

CHAPTER 2

INTRODUCTION OF DESIGNER TOOLS

2.1. INTRODUCTION TO ARDUINO IDE

The Arduino IDE (Integrated Development Environment) is an open-source software platform that provides a user-friendly interface for writing and uploading code to Arduino boards. It is designed to be easy to use for beginners yet powerful enough for experienced programmers. The Arduino IDE is available for free on Windows, macOS, and Linux operating systems.

Selecting the Arduino Board:

To avoid any error while uploading your program to the board, you must select the correct Arduino board name, which matches with the board connected to your computer.

Go to Tools → Board and select your board. For most of example illustrations, we are using an Arduino Uno board, but you must select the name, and sometime also the version, matching the board that you are using: Arduino Nano, Arduino Pro Mini 3.3v and 8MHz,...

For INTEL-IRRIS training & capacity sessions, we designed a PCB board where an Arduino Pro Mini microcontroller board can be plugged. In this case select Arduino Pro Mini in Arduino AVR boards section as the target board. Also select ATmega328P (3.3V, 8 MHz) as processor version.

Selecting the Serial Port:

Go to Tools → Port menu. This is likely /dev/ttyUSB0 on Linux machine, or /dev/cu.usbserialXXXXX on MacOS computers or COM3 or higher (COM1 and COM2 are usually reserved for hardware serial ports) on Windows computers.

To find out, you can disconnect your Arduino board and re-open the menu, the entry that disappears should be of the Arduino board. Reconnect the board and select that serial port.

It is very common now to have Arduino-compatible boards coming from Chinese manufacturers. In most cases, you will need the CH341 drivers to get your serial port detected. Here are the links to CH341 drivers for various OS if your board is not detected. Install the required driver, restart your computer and re-launch your Arduino IDE to see the serial port for your Arduino board.

- Windows
- Linux
- MacOS

Uploading the program to your board:

Simply click the Upload button in the Arduino IDE window. Wait a few seconds, you will see the RX and TX LEDs on the board flashing. If the upload is successful, the message Done uploading will appear in the status bar.

If you have issue uploading your sketch, such as out-of-sync message or programmer not responding message, check your board type and also avoid connecting the board to your computer with a too long USB cable. Also avoid connecting through a USB hub.

Output:

You can click on the Serial Monitor button to open the serial monitor in order to show the output from the Arduino board (whatever is written with the `Serial.print()` function).

Be sure to select at the bottom of the Serial Monitor window, the baud rate corresponding to what has been set in your program (in the example here it is 9600).

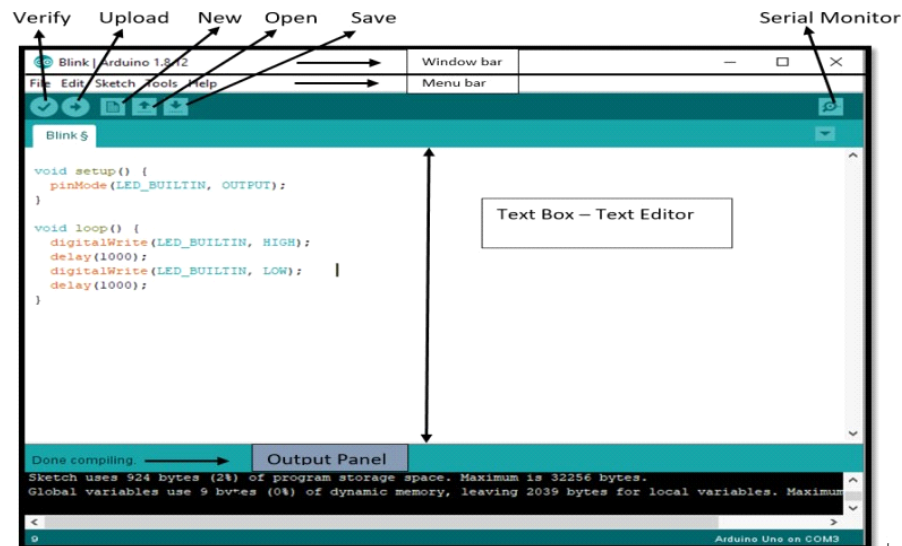


Figure 2.1: Arduino IDE Software

The key features for Arduino IDE are:

- **Code Editor:** It is a text editor that is specifically designed for writing Arduino code. This Editor provides syntax highlighting auto-completion, and other features.
- **Compiler:** It is used to convert Arduino code into machine code and it also checks for syntax errors and provide warnings.
- **Uploader:** The Arduino IDE includes an uploader that uploads the compiled code to Arduino board.
- **Serial Monitor:** This serial monitor will allow you to communicate with your Arduino board. It is used to send and receive data from the board .

2.2 INTRODUCTION TO PROTEUS SOFTWARE

Proteus Design Suite is a powerful electronic design automation (EDA) software that helps engineers and hobbyists design, simulate, and layout printed circuit boards (PCBs). It is a versatile tool that can be used for a wide range of projects, from simple circuits to complex microcontroller systems. The Benefits of using proteus Design suite are reducing design time, Improves design quality, saves money and increase in productivity.

Proteus Design Suite is a valuable tool for anyone who designs and builds electronic circuits. It is a versatile and powerful tool that can help to create high-quality designs

in a short amount of time. The key Features of proteus design suite are Schematic capture, Simulation, PCB layout design, 3D board visualization, Microcontroller simulation, IDE for simulation and lot more.

- Proteus is used to simulate, design, and drawing of electronic circuits. It was invented by Labcenter Electronics.
- By using proteus you can make two-dimensional circuit designs as well.
- With the use of this engineering software, you can construct and simulate different electrical and electronic circuits on your personal computers or laptops. Designing of circuits on the Proteus takes less time than practical construction of the circuit.
- The possibility of error is less in software simulation such as loose connection that takes a lot of time to find out connections problems in a practical circuit.
- Circuit simulations provide the main feature that some components of circuits are not practical then you can construct your circuit on Proteus.
- There is zero possibility of burning and damaging of any electronic component in Proteus.
- Electronic tools that are very expensive can easily get in Proteus such as an oscilloscope.
- Using Proteus you can find different parents of circuits such as current, the voltage value of any component, and resistance at any instant which is very difficult in a practical circuit.

There are numerous benefits to simulating circuits on Proteus before making them practically.

Steps to make circuit in Proteus:

- First of all click on Proteus Icon in your computer and click on a new file option.
- After that, you will see the drawing sheet and Save it according to your project.
- After a move to the component option and select the elements for your projects.
- After clicking on components mode you will see two buttons P and L. If you move to P button you will see Pic from Libraries.
- It is used to select different components for circuit construction.
- When you will click on the P button you will see. Type your component for a circuit.
- When you will select components for your project you will see them in a box.

- I have also selected some components for designing of a simple circuit.
- After the selection of components make the circuit layout of your project and connect all these components with the wires.
- For connection of one component to another clicks the left of the first terminal of the component and drag it to other components.
- If you want to remove any component or remove its connection just double-click on the respective of component of the wire.
- If you want to change the values of any component such as resistance, or capacitor, then click right on that component and select the desired value and click OK button.
- As in the below figure, I vary the value of battery voltages.
- When you connect all components in the circuit like run button in the left bottom see the practical working of your circuit.
- When you will observe the simulation of your circuit than click on stop button on the left bottom to stop the working of the circuit.

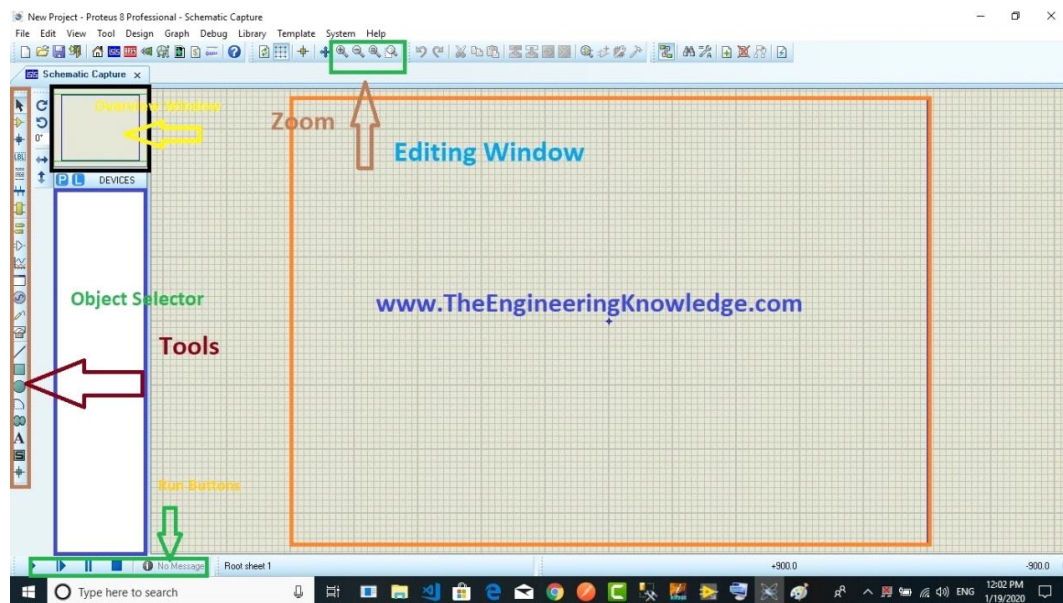


Figure 2.2 Proteus Software

CHAPTER 3

INTRODUCTION OF HARDWARE COMPONENTS

3.1. ARDUINO UNO

Arduino boards are generally based on microcontrollers from Atmel Corporation like 8, 16 or 32 bit AVR architecture based microcontrollers.

Arduino is an open-source prototyping platform in electronics based on easy-to-use hardware and software. Subtly speaking, Arduino is a microcontroller based prototyping board which can be used in developing digital devices that can read inputs like finger on a button, touch on a screen, light on a sensor etc. and turning it in to output like switching on an LED, rotating a motor, playing songs through a speaker etc

Arduino boards are available in different sizes, form factors, different no. of I/O pins etc. Some of the commonly known and frequently used Arduino boards are Arduino UNO, Arduino Mega, Arduino Nano, Arduino Micro and Arduino Lilypad.

Arduino UNO is a microcontroller board based on the ATmega328P. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started. You can tinker with your UNO without worrying too much about doing something wrong, worst case scenario you can replace the chip for a few dollars and start over again.

The name 'Uno' was chosen to commemorate the first release of Arduino software in Italy (IDE). Arduino Uno is an 8-bit ATmega328P microcontroller-based microcontroller board. It includes a crystal oscillator, serial communication, voltage regulator, and other components to support the ATmega328P microprocessor. The Arduino Uno contains 14 digital input/output pins, 6 analog input pins, a USB connection, a Power barrel connector, an ICSP header, and a reset button.

3.1.1. TECHNICAL SPECIFICATIONS:

- Microcontroller : Microchip ATmega328P
- Flash Memory : 32KB of which 0.5KB used by boot loader
- SRAM : 2KB
- EEPROM : 1KB
- Analog Input Pins : 6
- DC Current Per I/O Pin : 20mA
- DC Current For 3.3V Pin : 50mA
- Operating Voltage : 5V

General Pin Configuration:

Vin: This is the input voltage pin of the Arduino board used to provide input supply from an external power source.

5V: This pin of the Arduino board is used as a regulated power supply voltage and it is used to give supply to the board as well as onboard components.

V: This pin of the board is used to provide a supply of 3.3V which is generated from a voltage regulator on the board

GND: This pin of the board is used to ground the Arduino board.

Reset: This pin of the board is used to reset the microcontroller. It is used to Resets the microcontroller.

Analog Pins: The pins A0 to A5 are used as an analog input and it is in the range of 0-5V.

Digital Pins: The pins 0 to 13 are used as a digital input or output for the Arduino board.

Serial Pins: These pins are also known as a UART pin. It is used for communication between the Arduino board and a computer or other devices. The transmitter pin number 1 and receiver pin number 0 is used to transmit and receive the data respectively.

External Interrupt Pins: This pin of the Arduino board is used to produce the External interrupt and it is done by pin numbers 2 and 3.

PWM Pins: This pins of the board is used to convert the digital signal into an analog by varying the width of the Pulse. The pin numbers 3,5,6,9,10 and 11 are used as a PWM pin.

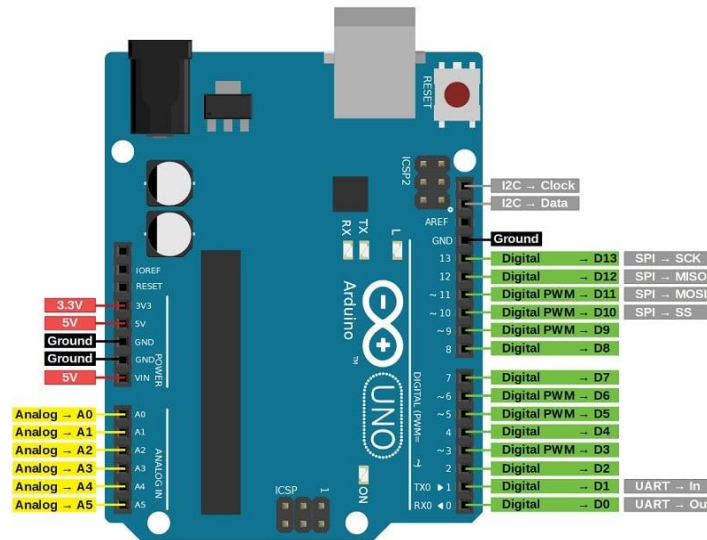


Figure 3.1,1. Pin Diagram of Arduino Uno Board

Some of the digital I/O pins have special functions which are describe below.

- Pins 0 and 1 are used for serial communication. They are used to receive and transmit serial data which can be used in several ways like programming the Arduino board and communicating with the user through serial monitor.
- Pins 2 and 3 are used for external interrupts. An external event can be triggered using these pins by detecting low value, change in value or falling or rising edge on a signal.
- As mentioned earlier, 6 of the 14 digital I/O Pins i.e. 3, 5, 6, 9, 10, and 11 can provide 8-bit PWM output.
- Pins 10, 11, 12 and 13 (SS, MOSI, MISO AND SCK respectively) are used for SPI communication.

- Pin 13 has a built-in LED connected to it. When the pin is HIGH, the LED is turned on and when the pin is LOW, it is turned off.

Arduino Uno has 6 analog input pins which can provide 10 bits of resolution i.e. 1024 different values. The analog pins on the Arduino UNO are labelled A0 to A5.

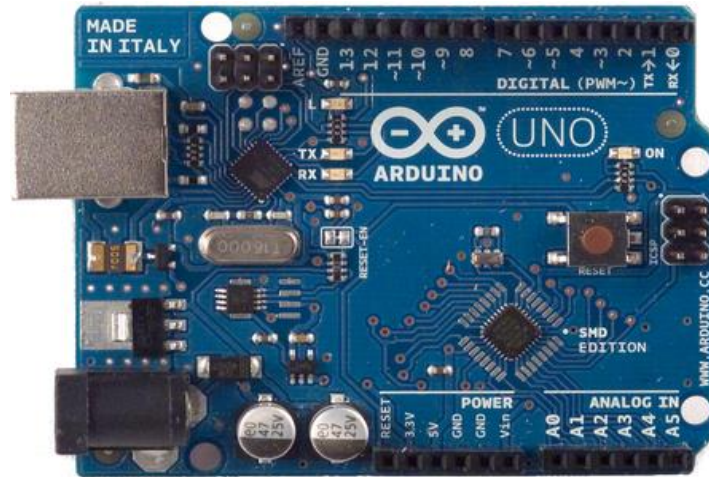


Figure 3.1.2. Arduino Uno Board

3.2. FINGERPRINT SENSOR

There are different types of fingerprint modules available in the market like R305, R307. For a better understanding of this sensor, here we are going to discuss an overview of R307 fingerprint sensor module.

R307-fingerprint-sensor-module

The R307 is one kind of fingerprint sensor module used in biometrics for security in fingerprint detection as well as verification. These devices are mainly used in safes where there is a high-powered DSP chip used in the rendering of image, feature-finding, searching and calculation by connecting it to any microcontroller with the help of TTL serial, & send data packets to get photos, notice prints, search and hash. The enrollment of new fingers can be stored directly within the flash memory of on board.

The working principle of the fingerprint sensor mainly depends on the processing. The fingerprint processing mainly includes two elements namely enrollment and matching.

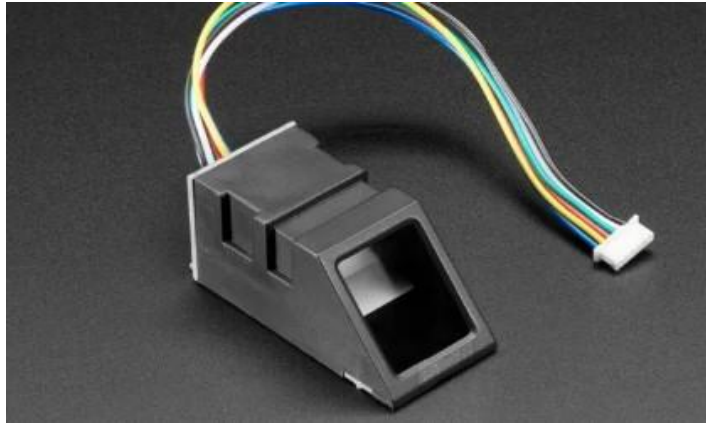


Figure 3.2. R307 Fingerprint Sensor

The method of capturing fingerprint data depends on the type of scanner being used:

Optical Sensor: These types of scanners basically make a photocopy of the finger. Many illuminate the finger in order to deliver crisp contrast of lines as the light-sensitive scanner (usually an image sensor or light-sensitive microchip) records the information to produce a digital image. Many PC-connected fingerprint scanners use optical sensors.

Capacitive Sensor: Instead of light, capacitive scanners use electricity (think of the way touchscreens work) to determine fingerprint patterns. As a finger rests on the touch-capacitive surface, the device measures the charge; ridges exhibit a change in capacitance, while valleys produce practically no change at all. The sensor uses all this data to accurately map out prints. Most all smartphones with fingerprint scanners use capacitive sensors.

Ultrasonic Sensor: Similar to how bats and dolphins use echolocation to find and identify objects, ultrasonic scanners work via sound waves. The hardware is designed to send out ultrasonic pulses and measure how much bounces back. Ridges and valleys reflect sound differently, which is how ultrasonic scanners are able to create a detailed 3D map of fingerprint patterns. Ultrasonic sensors are currently being prototyped (e.g. by Qualcomm Technologies, Inc.) and tested for use in mobile devices.

3.3. RELAY MODULE

Relay modules are simply circuit boards that house one or more relays. They come in a variety of shapes and sizes, but are most commonly rectangular with 2, 4, or 8 relays mounted on them, sometimes even up to a 16 relays.

A relay is an electrical switch that can be used to control devices and systems that use higher voltages. In the case of module relay, the mechanism is typically an electromagnet.

A relay module is available in an array of input voltage ratings: It can be a 3.2V or 5V relay module for low power switching, or it can be a 12 or 24V relay module for heavy-duty systems.

The relay module information is normally printed on the surface of the device for ready reference. This includes the input voltage rating, switch voltage, and current limit.



Figure 3.3. Relay Module

3.4. BUZZER

Buzzers are electric sounding devices that generate sounds. Typically powered by DC voltage, they can be categorised as Piezo buzzer and magnetic buzzer.

An electromagnetic buzzer consists of an oscillator, solenoid coil, magnet, vibration diaphragm, housing etc. Pretty much works the same as a magnetic buzzer, where they produce sound through magnetism, with a frequency of 2 kHz.



Figure 3.4. Buzzer

The piezoelectric type uses the piezoelectric ceramic's piezoelectric effect & pulse current to make the metal plate vibrate & generate sound. This kind of buzzer is made with a resonance box, multi resonator, piezoelectric plate, housing, impedance matcher, etc. Some of the buzzers are also designed with LEDs.

Types of Buzzer:-

Electromechanical:

Early devices were based on an electromechanical system identical to an electric bell without the metal gong. Similarly, a relay may be connected to interrupt its own actuating current, causing the contacts to buzz (the contacts buzz at line frequency if powered by alternating current) Often these units were anchored to a wall or ceiling to use it as a sounding board. The word "buzzer" comes from the rasping noise that electromechanical buzzers made.

Mechanical:

A joy buzzer is an example of a purely mechanical buzzer and they require drivers. Other examples of them are doorbells

Piezoelectric:

A piezoelectric element may be driven by an oscillating electronic circuit or other audio signal source, driven with a piezoelectric audio amplifier. Sounds commonly used to indicate that a button has been pressed are a click, a ring or a beep. Interior of a readymade loudspeaker, showing a piezoelectric-disk-beeper (With 3 electrodes ...

including 1 feedback-electrode the central, small electrode joined with red wire in this photo), and an oscillator to self-drive the buzzer. A piezoelectric buzzer/beeper also depends on acoustic cavity resonance or Helmholtz resonance to produce an audible beep.

3.5. BATTERY

A battery is a device that converts chemical energy to electrical energy. A battery's chemical reactions involve the flow of electrons from one material (electrode) to another via an external circuit. The flow of electrons generates an electric current, which can be used to perform work.

Types of Batteries

Batteries are classified into-

I. Primary battery/cell

II. Secondary battery/ cell

Primary battery/cell

These batteries cannot be recharged once they have been used because the devices are not easily reversible, and active materials may not return to their original forms.

Secondary battery/cell

I. Cylindrical Cells: Cylindrical cells are the most widely used battery packs for both primary and secondary batteries, and they are also among the simplest to produce. These cells contain a pressure relief mechanism that allows them to withstand high internal pressure without deforming.

An electric battery is a source of electric power consisting of one or more electrochemical cells of external connections for powering electrical devices.

When a battery is supplying power, its positive terminal is the cathode and its negative terminal is the anode. The terminal marked negative is the source of electrons that will flow through an external electric circuit to the positive terminal. When a battery is connected to an external electric load, a redox reaction converts high-energy reactants to lower-energy products, and the free-energy difference is

delivered to the external circuit as electrical energy. Historically the term "battery" specifically referred to a device composed of multiple cells; however, the usage has evolved to include devices composed of a single cell.

Primary (single-use or "disposable") batteries are used once and discarded, as the electrode materials are irreversibly changed during discharge; a common example is the alkaline battery used for flashlights and a multitude of portable electronic devices.

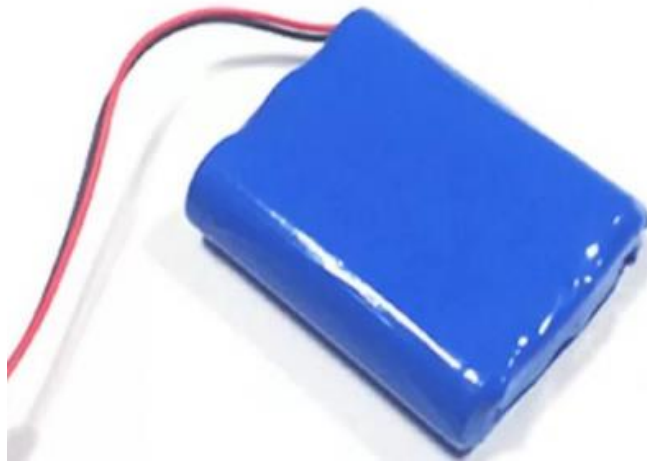


Figure 3.5. Rechargeable Batteries

3.6. JUMPER CABLES

A jump wire is an electrical wire, or group of them in a cable, with a connector or pin at each end (or sometimes without them, simply "tinned"), which is normally used to interconnect the components of a breadboard or other prototype or test circuit, internally or with other equipment or components, without soldering.

Individual jump wires are fitted by inserting their "end connectors" into the slots provided in a breadboard, the header connector of a circuit board, or a piece of test equipment.



Figure 3.6. Jumper Cables

Jumper wires of insulated 26AWG wire terminated with crimped pins or sockets in plastic housing are used to make connections between pin headers or sockets. With 2.5 mm (0.1 inch) housing they're suitable for interfacing single board computers like the Arduino and Raspberry Pi.

CHAPTER 4

IMPLEMENTATION OF HARDWARE AND SOFTWARE COMPONENTS

4.1. WORKING

The motorcycle will be ignited only when the authorized person scans his or her finger on the fingerprint sensor module. The finger prints of the authorized persons are stored in the fingerprint module. When any person puts his or her finger on the fingerprint module, the data of the placed finger is checked to see if it matches the stored data in the module. If the fingerprint data is found in the module, then the microcontroller in the Arduino UNO ignites or starts the bike through a relay; otherwise, the bike will emit a beep sound produced by the buzzer. The Arduino board is powered by a battery.

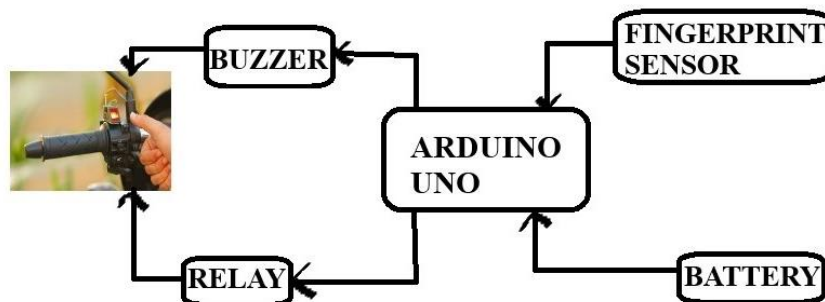


Figure 4.1. Block Diagram of Proposed Model

4.2 . FLOWCHART

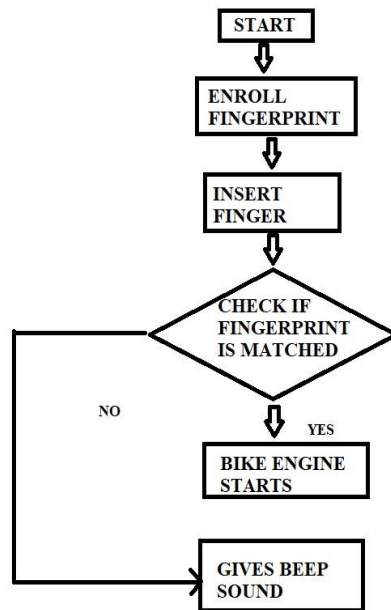


Figure 4.2. Flowchart

The flow chart tells us the step-by-step procedure.

1. Power on the system, which includes the microcontroller, sensors, and other
2. Initialize the system, which consists of fingerprint sensors.
3. Read the system configuration file, which means to read the instructions from the configuration file and the system operation according to the configuration file.
4. Read data from the fingerprint sensors and analyze the data to check whether it is required to start the ignition or not.
5. If the data doesn't match, initialize the buzzer and a beep sound.

4.3. PROGRAM REPRESENTATION

4.3.1. SOURCE CODE FOR ENROLLING FINGERPRINT

```
#include <Adafruit_Fingerprint.h>

SoftwareSerial mySerial(2, 3);

Adafruit_Fingerprint finger = Adafruit_Fingerprint(&mySerial);

uint8_t id;

void setup()
{
  Serial.begin(9600);
  while (!Serial); // For Yun/Leo/Micro/Zero/...
  delay(100);
  Serial.println("\n\nAdafruit Fingerprint sensor enrollment");

  // set the data rate for the sensor serial port
  finger.begin(57600);

  if (finger.verifyPassword()) {
    Serial.println("Found fingerprint sensor!");
  } else {
    Serial.println("Did not find fingerprint sensor :(");
    while (1) { delay(1); }
  }
}

uint8_t readnumber(void) {
  uint8_t num = 0;

  while (num == 0) {
    while (! Serial.available());
    num = Serial.parseInt();
  }
}
```

```

return num;
}

void loop() // run over and over again
{
  Serial.println("Ready to enroll a fingerprint!");
  Serial.println("Please type in the ID # (from 1 to 127) you want to save this finger
  as...");
  id = readnumber();
  if (id == 0) { // ID #0 not allowed, try again!
    return;
  }
  Serial.print("Enrolling ID #");
  Serial.println(id);

  while (! getFingerprintEnroll() );
}

uint8_t getFingerprintEnroll() {

  int p = -1;
  Serial.print("Waiting for valid finger to enroll as #"); Serial.println(id);
  while (p != FINGERPRINT_OK) {
    p = finger.getImage();
    switch (p) {
      case FINGERPRINT_OK:
        Serial.println("Image taken");
        break;
      case FINGERPRINT_NOFINGER:
        Serial.println(".");
        break;
      case FINGERPRINT_PACKETRECEIVEERR:
        Serial.println("Communication error");
        break;

```

```

case FINGERPRINT_IMAGEFAIL:
Serial.println("Imaging error");
break;
default:
Serial.println("Unknown error");
break;
}
}

// OK success!

p = finger.image2Tz(1);
switch (p) {
case FINGERPRINT_OK:
Serial.println("Image converted");
break;
case FINGERPRINT_IMAGEMESS:
Serial.println("Image too messy");
return p;
case FINGERPRINT_PACKETRECEIVEERR:
Serial.println("Communication error");
return p;
case FINGERPRINT_FEATUREFAIL:
Serial.println("Could not find fingerprint features");
return p;
case FINGERPRINT_INVALIDIMAGE:
Serial.println("Could not find fingerprint features");
return p;
default:
Serial.println("Unknown error");
return p;
}

Serial.println("Remove finger");

```

```

delay(2000);
p = 0;
while (p != FINGERPRINT_NOFINGER) {
p = finger.getImage();
}
Serial.print("ID "); Serial.println(id);
p = -1;
Serial.println("Place same finger again");
while (p != FINGERPRINT_OK) {
p = finger.getImage();
switch (p) {
case FINGERPRINT_OK:
Serial.println("Image taken");
break;
case FINGERPRINT_NOFINGER:
Serial.print(".");
break;
case FINGERPRINT_PACKETRECEIVEERR:
Serial.println("Communication error");
break;
case FINGERPRINT_IMAGEFAIL:
Serial.println("Imaging error");
break;
default:
Serial.println("Unknown error");
break;
}
}

// OK success!

p = finger.image2Tz(2);
switch (p) {
case FINGERPRINT_OK:

```

```

Serial.println("Image converted");
break;
case FINGERPRINT_IMAGEMESS:
Serial.println("Image too messy");
return p;
case FINGERPRINT_PACKETRECIEVEERR:
Serial.println("Communication error");
return p;
case FINGERPRINT_FEATUREFAIL:
Serial.println("Could not find fingerprint features");
return p;
case FINGERPRINT_INVALIDIMAGE:
Serial.println("Could not find fingerprint features");
return p;
default:
Serial.println("Unknown error");
return p;
}

// OK converted!
Serial.print("Creating model for #"); Serial.println(id);

p = finger.createModel();
if (p == FINGERPRINT_OK) {
Serial.println("Prints matched!");
} else if (p == FINGERPRINT_PACKETRECIEVEERR) {
Serial.println("Communication error");
return p;
} else if (p == FINGERPRINT_ENROLLMISMATCH) {
Serial.println("Fingerprints did not match");
return p;
} else {
Serial.println("Unknown error");
return p;
}

```

```

}

Serial.print("ID "); Serial.println(id);
p = finger.storeModel(id);
if (p == FINGERPRINT_OK) {
  Serial.println("Stored!");
} else if (p == FINGERPRINT_PACKETRECEIVEERR) {
  Serial.println("Communication error");
  return p;
} else if (p == FINGERPRINT_BADLOCATION) {
  Serial.println("Could not store in that location");
  return p;
} else if (p == FINGERPRINT_FLASHERR) {
  Serial.println("Error writing to flash");
  return p;
} else {
  Serial.println("Unknown error");
  return p;
}
}

```


4.3.2 SOURCE CODE FOR MOTORCYCLE START USING FINGERPRINT

```
#include <Adafruit_Fingerprint.h>

SoftwareSerial mySerial(2, 3);

Adafruit_Fingerprint finger = Adafruit_Fingerprint(&mySerial);

void setup()
{
  pinMode (8,OUTPUT);
  pinMode (11,OUTPUT);
  Serial.begin(9600);
  while (!Serial); // For Yun/Leo/Micro/Zero/...
  delay(100);
  Serial.println("\n\nAdafruit finger detect test");

  // set the data rate for the sensor serial port
  finger.begin(57600);

  if (finger.verifyPassword()) {
    Serial.println("Found fingerprint sensor!");
  } else {
    Serial.println("Did not find fingerprint sensor :(");
    while (1) { delay(1); }
  }

  finger.getTemplateCount();
  Serial.print("Sensor contains "); Serial.print(finger.templateCount); Serial.println("
  templates");
  Serial.println("Waiting for valid finger...");
}

void loop() // run over and over again
{
```

```

getFingerprintIDez();
delay(50); //don't ned to run this at full speed.
}

```

```

uint8_t getFingerprintID() {
uint8_t p = finger.getImage();
switch (p) {
case FINGERPRINT_OK:
Serial.println("Image taken");
break;
case FINGERPRINT_NOFINGER:
Serial.println("No finger detected");
return p;
case FINGERPRINT_PACKETRECIEVEERR:
Serial.println("Communication error");
return p;
case FINGERPRINT_IMAGEFAIL:
Serial.println("Imaging error");
return p;
default:
Serial.println("Unknown error");
return p;
}
}

```

```

// OK success!

```

```

p = finger.image2Tz();
switch (p) {
case FINGERPRINT_OK:
Serial.println("Image converted");
break;
case FINGERPRINT_IMAGEMESS:
Serial.println("Image too messy");
return p;
}

```

```

case FINGERPRINT_PACKETRECEIVEERR:
Serial.println("Communication error");
return p;
case FINGERPRINT_FEATUREFAIL:
Serial.println("Could not find fingerprint features");
return p;
case FINGERPRINT_INVALIDIMAGE:
Serial.println("Could not find fingerprint features");
return p;
default:
Serial.println("Unknown error");
return p;
}

// OK converted!
p = finger.fingerFastSearch();
if (p == FINGERPRINT_OK) {
Serial.println("Found a print match!");
} else if (p == FINGERPRINT_PACKETRECEIVEERR) {
Serial.println("Communication error");
return p;
} else if (p == FINGERPRINT_NOTFOUND) {
Serial.println("Did not find a match");
return p;
} else {
Serial.println("Unknown error");
return p;
}

// found a match!
Serial.print("Found ID #"); Serial.print(finger.fingerID);
Serial.print(" with confidence of "); Serial.println(finger.confidence);

return finger.fingerID;

```

```

}

// returns -1 if failed, otherwise returns ID #
int getFingerprintIDez() {
  uint8_t p = finger.getImage();
  if (p != FINGERPRINT_OK) return -1;

  p = finger.image2Tz();
  if (p != FINGERPRINT_OK) return -1;

  p = finger.fingerFastSearch();
  if (p != FINGERPRINT_OK)
  {
    Serial.println("Finger is not valid");
    digitalWrite(11,HIGH);
    delay(1500);
    digitalWrite(11,LOW);
    return -1;
  }

  // found a match!
  digitalWrite(8,HIGH);
  delay(1500);
  digitalWrite(8,LOW);

  Serial.print("Found ID #"); Serial.print(finger.fingerID);
  Serial.print(" with confidence of "); Serial.println(finger.confidence);
  return finger.fingerID;
}

```

4.4. CIRCUIT CONNECTION PROCEDURE

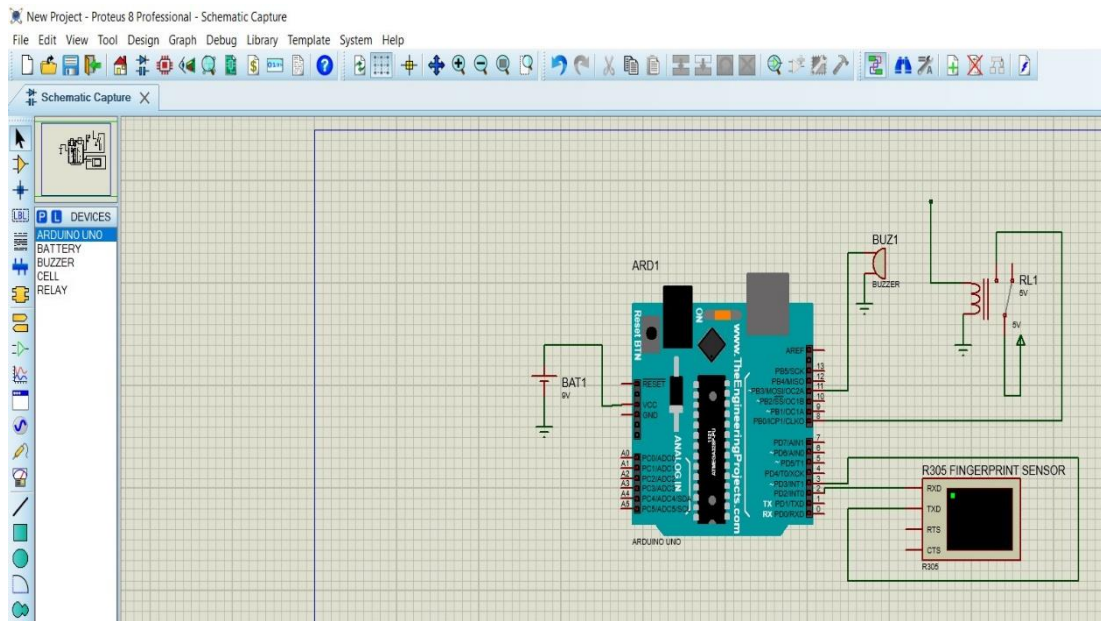


Figure 4.4. Circuit Implementation in Proteus Software

CHAPTER 5

CONCLUSION

HOW DOES IT WORKS

Arduino consists of both a physical programmable circuit board (often referred to as a microcontroller) and a piece of software, or IDE (Integrated Development Environment), that runs on your computer and is used to write and upload computer code to the physical board. Checks for moisture content; if there is moisture, then the pump will not do anything.

5.1 ENROLLING FINGERPRINT USING SERIAL MONITOR

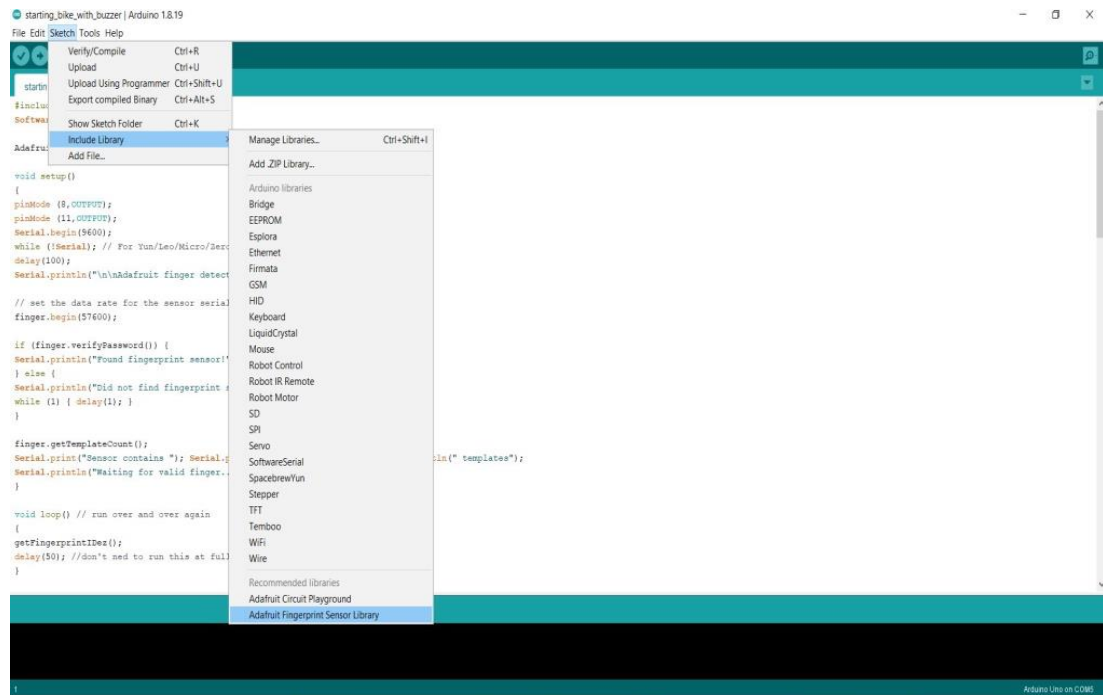


Figure 5.1.1. Adding Adafruit Fingerprint Library

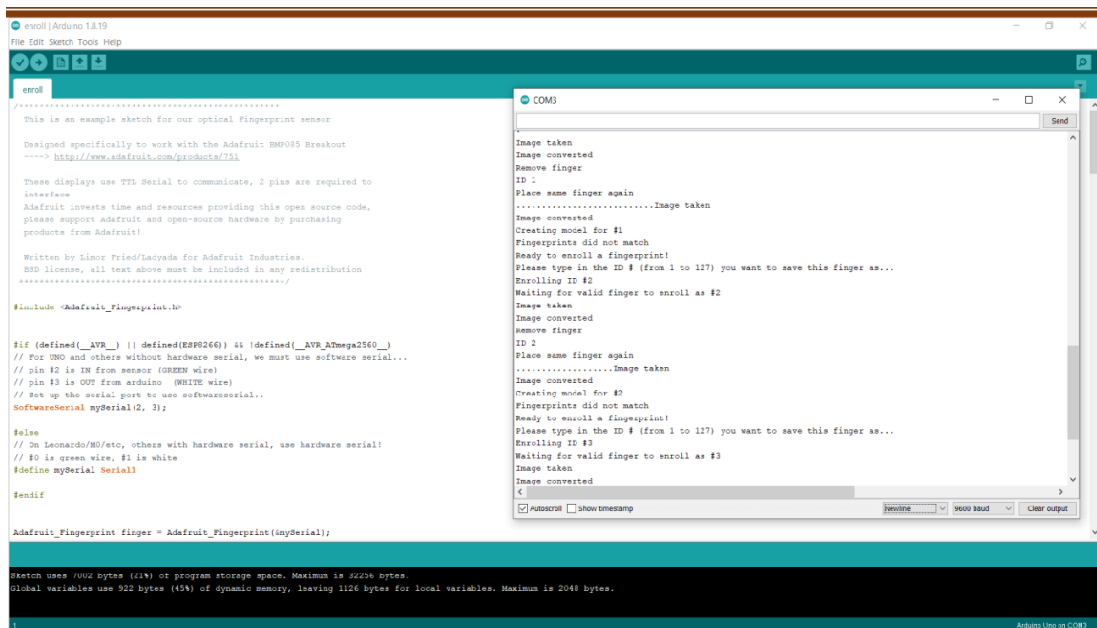


Figure 5.1.2. Enrolling the Fingerprint using Serial Monitor

5.2 DETECTION OF VALID OR INVALID FINGERPRINT

For detecting the fingerprint, whether it is enrolled or not, we check using the serial monitor by placing the finger on the fingerprint sensor. If the placed finger is enrolled or matched, it makes the ignition on by activating the relay module indicated by giving a green light; otherwise, if the finger is not matched, it will make the relay on by giving a red light, which will activate the buzzer.

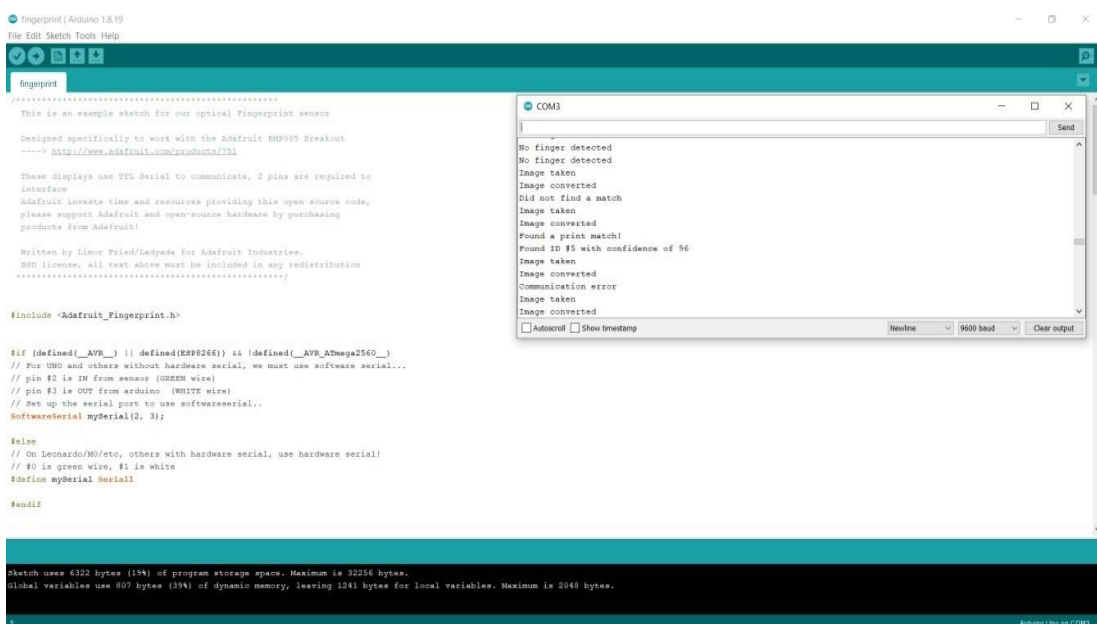


Figure 5.2.Detection of Valid /Invalid Fingerprint using Serial Monitor

A biometric system improves the security of a vehicle and makes it easier to start the bike only for authorized people whose fingerprints are stored in the memory. It does not allow unauthorized persons to start the bike and produces the buzzer sound when they try to access it. In this paper, we have implemented a low-cost and easily available system on a vehicle that provides more security than normal locks. This implementation of two-wheeler vehicle security can be extended to four-wheeler vehicles by using the improved version.

REFERENCES

1. Kiran CS. Anti-theft Fingerprint Security System for Motorcycles Using Arduino Uno, GSM, and GPS Modules In: Indian Journal of Science and Technology, Vol. 12(42), DOI:10.17485/ijst/2019/v12i42/145823, November 2019.
2. Wimberly H., Liebrock LM. Using fingerprint authentication for system security: an empirical study In: IEEE Symposium on Security and Privacy, 2011. P. 1–9.
3. Ega Supriatna, Adam Faroqi, Edi Mulyana, and Gilang Aditya Purba Motorcycle Security System Using Fingerprint and Voice Commands Based on Microcontrollers IEEE 2022 8th International Conference Paper
4. Verma P., Bhatia JS. Design and development of a GPS-GSM-based tracking system with Google Maps-based monitoring. Int J Comput Sci Eng Appl (IJCSEA). 2013;3(3):1–12.
5. Singh P, Sethi T, Biswal BB, and Pattanayak SK A smart anti-theft system for vehicle security Int J Mater Mech Manuf. 2015;3(4):1–15.
6. Alshamsi H, Kėpuska V, Alshamsi H. Real-time vehicle tracking using Arduino Mega. Int J Sci Technol. 2016;5(12):1–4.